Associations Between Skeletal Muscle Mass, Fat-Free Mass, and Saliva Osmolality: A Cross-Sectional Study

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Abstract

Objective: This investigation evaluated the relationships between skeletal muscle mass (SMM), fat-free mass (FFM), and hydration status measured through saliva osmolality (SOSM). It was hypothesized that SMM and FFM would positively associate with SOSM. **Design:** A cross-sectional, observational study. **Setting:** Laboratory and field collection. **Participants:** Seventy-two adults (42 men, 30 women; 29.3 ± 10.6 years, 171 ± 9 cm, 78.8 ± 15.0 kg, 28.7% ± 8.1% body fat). **Independent variables:** Skeletal muscle mass and FFM were measured through bioelectrical impedance analysis. **Outcomes:** Saliva osmolality was measured using the MX3 Hydration Testing System, a portable, noninvasive osmometer. Participants presented 2 saliva samples 3 to 5 minutes apart. Fluid intake was avoided for at least 5 minutes before sampling. **Results:** Linear regression analysis showed a nonsignificant association between SMM and SOSM (unstandardized B = 0.44, $P = 0.144$), as well as FFM and SOSM (unstandardized B = 0.23, $P = 0.172$). The adjusted analysis including covariates (age, sex, past 4-hour fluid intake, and time of sampling) showed nearly no effect, with insignificant associations between SMM and SOSM (unstandardized B = 0.98, $P = 0.146$) and FFM and SOSM (unstandardized B = 0.44, $P = 0.173$) remaining. Similarly, Spearman correlation coefficients between SMM and SOSM [$\rho(72) = 0.10$, $P = 0.427$] as well as FFM and SOSM [$\rho(72) = 0.08$ $P = 0.487$] were positive but nonsignificant and small. **Conclusions:** Despite growing evidence that urinary hydration biomarkers naturally vary with SMM and FFM, these 2 body composition variables seem to be insignificantly associated with hydration status according to SOSM. This may be related to relatively small concentrations of muscle metabolites (eg, creatinine, uric acid, urea) found in saliva.

Key Words: hydration, osmolality, saliva, creatinine, muscle metabolites

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INTRODUCTION

It is well-understood that adequate hydration status is required for normal human function.[1] Inadequate hydration status, also known as hypohydration, may bring about declines in physiological, cognitive, and physical functioning, particularly in thermally stressful environments.[2,3] For example, hypohydration can impair sweating, leading to detrimental effects on evaporative cooling and ultimately thermoregulation in the heat.[4] Indeed, a Joint Position Statement from the American College of Sports Medicine, Academy of Nutrition and Dietetics, and Dietitians of Canada states that exercise performance may become impaired when hypohydration exceeds 2% of body weight, with losses of 6% to 10% of body weight raising the risks of heat stroke and heat exhaustion.[2] Beyond athletic populations, hydration behaviors and biomarkers have also been linked with a variety of health conditions among the general population, including urinary tract infections, kidney stones, metabolic diseases such as type 2 diabetes, and changes in kidney function over time.[5] Given the importance of water and adequate hydration for optimizing physiological function, health, and performance, a plethora of hydration status assessment methods—urine specific gravity (USG), urine osmolality, plasma osmolality, total body water—have been used in research and clinical/field settings.

Studies have indicated that certain measures of hydration status, namely urine osmolality and USG, tend to increase with body mass and body mass index.[6–8] These results seem to be largely driven by variations in fat-free mass (FFM) and skeletal muscle mass (SMM).[8–11] From a physiological perspective, having large absolute amounts of SMM and FFM may contribute to elevated urine osmolality and specific gravity through increases in the excretion of creatinine, a breakdown byproduct of creatine from muscle. Other protein metabolites such as urea and uric acid may also be present in greater quantities in large, muscular people.[11] Ultimately, this means that individuals with more SMM and FFM may be at greater likelihood of being misdiagnosed as hypohydrated when they are not (ie, false-positive).[8–11]

Compared with classic hydration biomarkers such as USG and urine osmolality, relatively less research is available on saliva osmolality (SOSM), which represents the number of solute particles per kilogram of saliva.[12] In an analysis of several hydration biomarkers during active and passive dehydration, SOSM was found to be a sensitive, specific,

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and a valid measure of hydration status during active periods of hypohydration (ie, physical activity in the heat).[13] Other studies, both sport and clinical in nature, support the notion that SOSM may represent a viable marker of hydration status,[14,15] particularly when used alongside additional assessment methods, such as changes in body mass, urine osmolality, and USG. However, despite the potential of SOSM as a practical, convenient measure of hydration in field settings, it has yet to be widely assessed in conjunction with body composition. As such, it is currently unknown whether SOSM tends to naturally increase with greater amounts of SMM and FFM—as is the case with urine osmolality and USG.

This investigation's purpose was to explore and evaluate the relationships between SMM, FFM, and hydration status measured through SOSM. It was hypothesized that SMM and FFM would be positively associated with SOSM. Given the established relationships between urine markers of hydration and body composition, identifying similar associations between SOSM and body composition metrics would provide further evidence of the need to potentially reform current hydration status classification guidelines based on variations in body size and composition.

METHODS

Participants

Seventy-two individuals (42 men, 30 women; 29.3 ± 10.6 years, 171 ± 9 cm, 78.8 ± 15.0 kg, 28.7% ± 8.1% body fat percentage) volunteered to participate and had complete data for relevant variables. Fifty-five participants identified as White (76.4%), 6 as 2 or more races/ethnicities (8.3%), 5 as Black or African American (6.9%), 4 as Asian or Pacific Islander (5.5%), 1 as Hispanic or Latino (1.4%), and 1 as other race/ethnicity (1.4%).

Participants were recruited through convenience and snowball sampling through multiple avenues in the Hampton Roads, Virginia area. Only a limited number of inclusion criteria were applied to determine eligibility given the relative novelty of the research question. Specifically, individuals were eligible if they were at least 18 years of age, weighed under 330 pounds, were free of any implanted electrical device (ie, pacemaker), and were not pregnant. The latter 2 criteria represent contraindications to using the body composition device while 330 pounds is the maximal measurement weight of the device.

Ethics

The investigation's protocols and procedures were submitted to and approved by the Institutional Review Board at Old Dominion University (Reference #23-058). Before participation and data collection, potential participants went through an informed consent process in which the procedures, inclusion criteria, potential risks, and benefits were explained, and prospective participants were given the option to choose to either voluntarily participate or not. The study's procedures complied with the principles of the Declaration of Helsinki.

Procedures

A variety of locations were used for assessments given the portability of the saliva osmometer device involved in data collection; locations included the Human Performance Laboratory at Old Dominion University, as well as several local gyms and a weightlifting event. Procedures were not altered by the location of assessment and data collection.

Participants first completed a background questionnaire inquiring about age, sex, race/ethnicity, exercise habits, and total fluid intake over the past 4 hours. Prior research indicates that self-reported fluid intakes for periods of up to 24 hours correlate moderately-to-strongly with direct volume measurements and water turnover biomarkers.[16,17] Three participants with otherwise complete data failed to report fluid intake over the past 4 hours and were therefore excluded from the analysis (dropping to sample size from 75 to 72). Self-reported exercise level was evaluated through a single question that asked participants to indicate their typical exercise/physical activity habits as either sedentary, lightly active, moderately active, or highly active.

The MX3 Hydration Testing System (MX3 Diagnostics, Austin, TX; MX3 HTS) was used as the instrument to assess SOSM. Participants who had fluids with them were asked to refrain from drinking for at least 5 minutes before obtaining an initial saliva sample. This avoidance of fluid intake for at least 5 minutes was based on the manufacturer's operating instructions and is supported by research showing that SOSM is artificially altered within 1 minute of rinsing or ingesting fluid but returns to baseline levels within no more than 15 minutes.[18] To further reduce the potential impact of recent fluid intake, participants were asked to generate a fresh saliva sample after swallowing all existing saliva. They then presented the sample on their tongue with an open mouth, followed by a researcher tapping the tip of a test strip to the saliva. A second saliva sample was also taken 3 to 5 minutes later, after the measurement of height and body composition (described later). Previous work with the MX3 HTS in the investigators' laboratory found that 2 measurements taken within this timeframe show moderate-to-good reliability (ICC = 0.75).[19] In addition, to improve reliability and further reduce any potential impact of recent fluid intake, an average of the 2 SOSM values was used for all analyses. Participants were categorized as hydrated if SOSM ≤65.9 mOsm, mildly dehydrated if SOSM 66.0 to 100.0 mOsm, moderately dehydrated if 100.1 to 150.0 mOsm, or severely dehydrated if >150 mOsm.[20]

After the first SOSM measurement but before the second SOSM measurement, participants' heights were measured using a stadiometer, and body composition was evaluated using an OMRON Full Body Composition Monitor Scale (HBF-514C; Omron Healthcare, Inc, Lake Forrest, IL) that relies on bioelectrical impedance analysis. In a previous study, the OMRON HBF-516C, which is identical to the HBF-514C except for minor software differences, produced body fat percentage estimates that were fairly similar to a four-compartment model (28% vs 24.7%) while demonstrating 5.6% total error in cross-sectional validity, 1.9% total error in longitudinal validity, and strong test–retest reliability (ICC = 1.0).[21] Before the scan, participants wiped down the palms of their hands and soles of their feet with a moist wipe. Researchers inputted each participant's height, age, and sex into the HBF-514C, which was followed by a period of approximately 30 seconds in which participants stood on the platform and held on to an attached handle with arms out forward and extended. Variables of interest derived from the device were body mass (kg), SMM (kg), and body fat (kg). Fat-

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free mass was calculated by subtracting body fat in kilograms from body mass in kilograms.

Statistics

SPSS software (version 29, IBM, Armonk, NY) was used for the statistical analyses. The distribution of variables was assessed by examining histograms and Q–Q plots. The data for average SOSM had a slight right skew, likely related to the presence of a few values >100.0 mOsm. Fluid intake over the previous 4 hours had a significant right-skewed distribution, while SMM and FFM had bimodal distributions. Thus, values for these variables are reported descriptively as median (25th-75th percentile).

Ordinary least squares linear regression was performed to determine the associations between SMM and SOSM, as well as FFM and SOSM. Unadjusted models were run first, which were followed by multivariate regression that included age, past 4-hour fluid intake, sex, and time of day of sampling (PM vs AM). To ensure assumptions of regression were met, standardized residuals were plotted using histograms and Q–Q plots. Inspection of these plots revealed that there was a right skew to the distributions, with 4 participants having standardized residuals of >2.0. However, removal of these participants had minor effects on the regression output and did not change the statistical significance of any of the parameters in the models. Thus, the results for the full data set are reported. The Spearman correlation coefficients were also run and interpreted to measure the size of relationship between variables. A Mann–Whitney *U* test was used to assess whether fluid intake and SOSM differed between participants assessed in the morning versus afternoon/evening. A two-sided *P*-value ≤0.05 was used as the threshold for determining statistical significance. Deidentified raw data are available as a supplement (see **Data, Supplemental Digital Content 1**, http://links.lww.com/JSM/A547).

RESULTS

Among the 72 adults who participated, 26 described themselves as highly active (36.1%), 35 as moderately active (48.6%), 9 as lightly active (12.5%), and 2 as sedentary (2.8%). Twenty-eight participants (38.9%) were assessed in the morning, while 44 (61.1%) were assessed in the afternoon/evening. The median SOSM value was 63.0 (55.1-74.3) mOsm and did not differ ($P = 0.917$) by time of day [morning: 65.0 (49.9-75.8) mOsm; afternoon/evening: 63.0 (55.6-73.6) mOsm]. The median fluid intake over the 4 hours before assessment was 0.591 (0.355-1.042) L and did not vary significantly ($P = 0.772$) between participants assessed in the morning versus afternoon/evening [0.710 (0.362-0.946) versus 0.591 (0.355-1.153) L]. Forty-two participants were categorized as hydrated (58.3%), 27 as mildly dehydrated (37.5%), 3 as moderately dehydrated (4.2%), and none as severely dehydrated (0.0%). The median values for SMM and FFM were 27.8 (19.5-32.7) kg and 57.2 (44.5-66.1) kg, respectively.

The results of the linear regression analyses are summarized in Tables 1 and 2. There was no significant associations between SMM and SOSM (unstandardized B = 0.44, $P = 0.144$), as well as FFM and SOSM (unstandardized B = 0.23, $P = 0.172$). Figures 1 and 2 depict these relationships. The adjusted analyses including covariates showed no notable effects, with insignificant associations remaining between SMM and SOSM (unstandardized B = 0.98, $P = 0.146$) as well as FFM and SOSM (unstandardized B = 0.44, $P = 0.173$). The Spearman correlation analyses further indicated nonsignificant, small, positive relationships between SMM and SOSM [$\rho(72) = 0.10$, $P = 0.427$] as well as FFM and SOSM [$\rho(72]) = 0.08$, $P = 0.487$].

DISCUSSION

The goal of this study was to explore the relationships between SOSM and body composition, specifically focusing on the impact of SMM and FFM. Our hypothesis that SMM and FFM would be positively associated with SOSM was not supported, although we cannot rule out the possibility that there is a real, very small association (eg, $\rho = 0.1$) between these variables. Studies have indicated that some measures of hydration status, namely urine osmolality and USG, tend to increase with body mass and body mass index,[6–8] with the association being largely driven by variations in FFM and SMM.[8–11] Because urine was the body fluid tested in these studies, extrapolation to other body fluids such as saliva and tears should be performed cautiously. Our data suggest that the relationships between urine hydration markers and body composition that have been shown with relative consistency in other studies may not apply when saliva is the body fluid of interest.

The use of SOSM in field, clinical, and research settings is a relatively new development in comparison with many other traditional hydration biomarkers. Plasma osmolality, which is sometimes referred to as a gold standard for measuring hydration status, is aggressively regulated by the body and is less responsive to changes in fluid intake than urinary markers.[22] Similarly, in an experiment that increased fluid intake in people consuming less than 1.2 L/day and decreased

TABLE 1. Multiple Regression of SMM Predicting SOSM

		Unstandardized Coefficients				
		B (95% CI)	*SE*	Standardized Coefficient: β	*t*	*P*
Model 1	Skeletal muscle mass (kg)	0.44 (−0.16 to 1.04)	0.30	0.17	1.48	0.144
Model 2	Skeletal muscle mass (kg)	0.98 (−0.35 to 2.31)	0.67	0.38	1.47	0.146
	Age (y)	−0.28 (−0.68 to 0.13)	0.20	−0.17	−1.36	0.180
	Male sex	−8.43 (−26.54 to 9.69)	9.08	−0.24	0.93	0.357
	Past 4h fluid intake (L)	−5.81 (−13.16 to 1.55)	3.68	−0.20	−1.58	0.120
	Time of d (PM vs AM)	−0.38 (−9.54 to 8.79)	4.59	−0.01	−0.08	0.935

SE, standard error.

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TABLE 2. Multiple Regression of FFM Predicting SOSM

		Unstandardized Coefficients				
		B (95% CI)	*SE*	Standardized Coefficient: β	*t*	*P*
Model 1	Fat-free mass (kg)	0.23 (−0.10 to 0.57)	0.17	0.16	1.38	0.172
Model 2	Fat-free mass (kg)	0.44 (−0.20 to 1.08)	0.32	0.31	1.38	0.173
	Age (y)	−0.32 (−0.73 to 0.10)	0.21	−0.19	−1.53	0.131
	Male sex	−5.57 (−21.10 to 9.96)	7.78	−0.16	0.72	0.477
	Past 4h fluid intake (L)	−5.51 (−12.79 to 1.78)	3.65	−0.19	−1.51	0.136
	Time of d (PM vs AM)	−0.44 (−9.63 to 8.76)	4.60	−0.01	−0.10	0.925

SE, standard error.

fluid intake in people consuming >2.0 L/day, SOSM did not change over time in either group.[23] This tight regulation of SOSM against changes in fluid consumption may help to explain why self-reported fluid intake was not associated with SOSM within our regression models.

Research directly correlating the osmolality of urine to that of salvia is limited. Both urine and SOSM have been shown to correlate strongly with plasma osmolality during dehydration.[13] Furthermore, changes in SOSM during passive dehydration may be smaller as compared with urine osmolality, with larger SOSM changes observed during active dehydration.[24] Some studies have compared USG with SOSM, finding strong correlations in first-morning values,[25] and both being suitable for serial measurement during active and passive dehydration.[24]

Although the specific gravity and osmolality of body fluids are significantly correlated with one another, they are not the same. Solute number and concentration are important factors that influence both osmolality and specific gravity, but USG additionally considers solute mass, meaning that it is substantially affected by the presence of larger or smaller solutes.[26] Although USG can be used to reasonably predict urine osmolality, variations in urine components (eg, glucose, protein, ketones, albumin, creatinine) can significantly affect that relationship.[27] For example, creatinine, which has a higher molecular weight than many other common urine solutes, is likely to have greater effects on specific gravity than osmolality.[26]

One major suspected reason for SMM/FFM having a smaller association with SOSM than USG is related to the composition of saliva. Given a lower concentration of muscle metabolites in saliva (eg, creatinine, urea, uric acid), it is reasonable to speculate that SOSM is subject to less influence by SMM. This is supported by evidence from several studies that have measured the presence of these muscle metabolites in saliva and serum or plasma. In these studies, healthy control groups produced salivary creatinine levels of 11.1 ± 0.3 μmol/L[28] and 0.08 ± 0.03 mg/dL,[29] salivary uric acid levels of 206 ± 48 μmol/L,[28] and salivary urea levels of 12.59 ± 0.66 mg/dL[29]; notably, all these saliva concentrations were less than the concentrations found in plasma or serum.[28,29] As it relates to urine, population-representative data from the National Health and Nutrition Examination Survey indicate mean spot urine creatinine concentrations may range

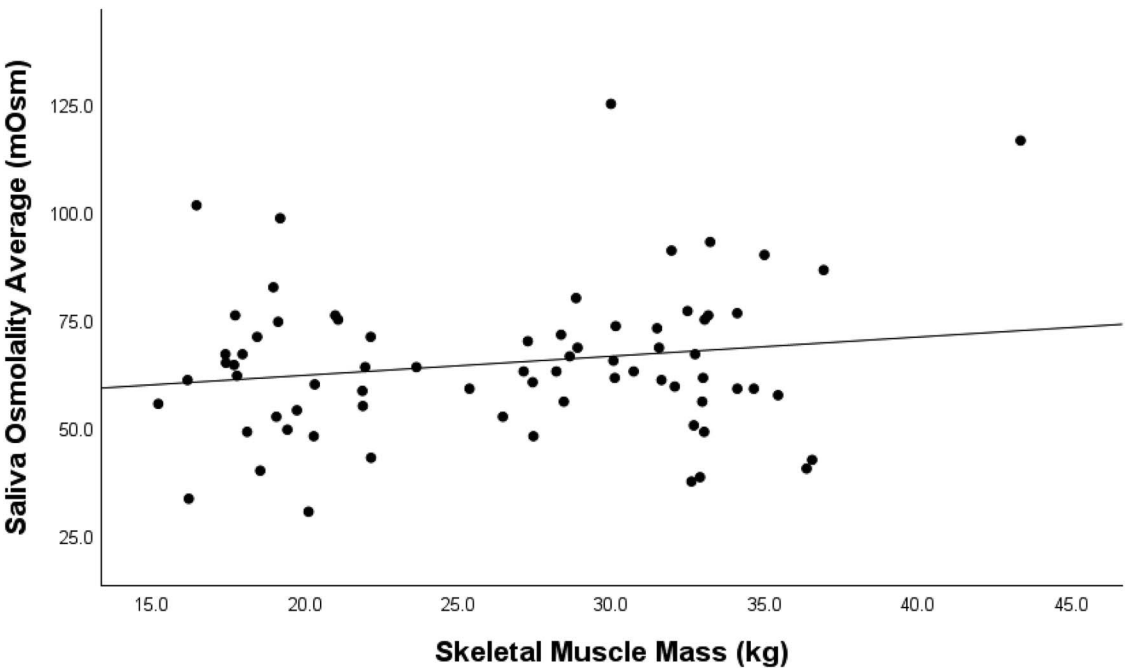

Figure 1. Scatterplot of the relationship between SMM and SOSM.

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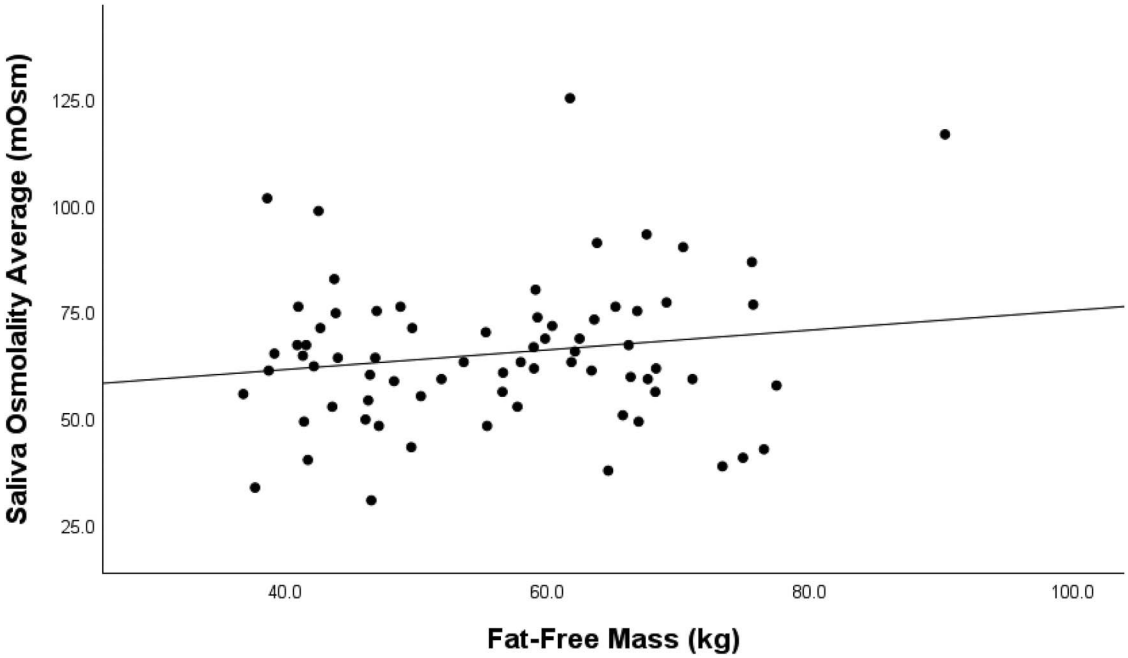

Figure 2. Scatterplot of the relationship between FFM and SOSM.

anywhere from 108 to 171 mg/dL in Black and White American adults.[30]

Limitations of this study include the imperfect validity of the HBF-514C for measuring body composition. In addition, acute changes in body water, as a result of either diet or exercise, can also affect body composition estimates and introduce error.[31] Despite these issues, the HBF-514C was primarily chosen because of portability issues with data collection outside of the laboratory. Although the accuracy of the SMM estimate from the HBF-514C is unknown, work by Siedler et al[21] has shown that this type of Omron device has relatively high validity for body fat percentage against a 4-compartment model (R^2 of 0.78, standard error of the estimate of 4.3%), which would also translate to relatively high validity for FFM. An additional limitation of this investigation is that relatively few exclusion criteria were used to determine participant eligibility, resulting in a heterogenous study sample, especially in relation to self-reported physical activity levels. In our view, this is justifiable given that this is the first study to examine the associations between SOSM and SMM, but future investigations should consider sampling more homogenous groups to reduce the influence of potentially confounding or moderating variables. Another study weakness is the lack of control over dietary intake leading up to study participation, specifically with fluid intake. It may have been ideal for the minimum period between last time of fluid consumption and saliva sampling to be longer than 5 minutes.[25] Furthermore, if water or a sports drink containing electrolytes and carbohydrates was consumed less than 15 minutes before SOSM measurement, then there is a chance SOSM was misclassified as very low or very high.[20] However, we used various strategies to mitigate the impact of recent fluid intake, including asking participants to swallow all existing saliva before generating a new sample and also taking a second sample 3 to 5 minutes after the initial sample. Despite one investigator assessing most of the saliva samples, all investigators used the MX3 HTS to measure SOSM at some point during the study, so between-investigator differences in exact sampling technique may have had some effect on SOSM measurements. That being said, an average of 2 SOSM values was used to reduce the influence of measurement error.

CONCLUSIONS

Despite growing evidence that SMM and FFM have an impact on hydration status measured through urine, these variables seem to be insignificantly associated with hydration status according to SOSM. This may be related to smaller concentrations of muscle metabolites (eg, creatinine, uric acid, urea) found in saliva. These results suggest that there is less of a need to take into consideration body size and body composition when assessing hydration status through SOSM. However, additional research that directly compares USG, urine osmolality, and SOSM while simultaneously measuring body composition with more sophisticated techniques is warranted. In addition, future research should aim to confirm the impact of muscle metabolites on SOSM, if any.

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